

Program : Diploma in Cloud computing and Big Data	
Course Code : 6279C	Course Title: Cloud Security Lab
Semester : 6	Credits: 1.5
Course Category: Programme Elective Course	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Learn how to secure compute services viz securing VMs, Managed Database Services and Securing serverless/function as a service
- Practice securing cloud based IAM services, conducting security monitoring and audit trails
- Experiment with securing Storage Services.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
How to use resources in public cloud	3276	Cloud Computing Lab	3
Basic security principles ,threat and management.		Cloud Security	6
Practical use of public cloud platforms providing IaaS, PaaS and SaaS services		Advanced cloud computing Lab	5

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Implement securing compute services	10	Applying
CO2	Experiment with securing networking services	9	Applying
CO3	Practice securing cloud based IAM services	10	Applying

CO4	Demonstrate security monitoring and audit trails and securing storage services	12	Applying
	Lab Test	4	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			3
CO2	3			3			3
CO3	3			3			3
CO4	3			3			3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Implement securing compute services		
M1.01	Configure Secure Amazon Elastic Compute Cloud (EC2)	4	Applying
M1.02	Implement security in Amazon RDS for MySQL	3	Applying
M1.03	Demonstrate securing AWS Lambda	3	Applying
CO2	Experiment with securing networking services		
M2.01	Experiment how to do security in Amazon Virtual Private Cloud	6	Applying
M2.02	Implement Secure Content Delivery Network Services	3	Applying

	Lab Test 1	2	
CO3	Practice securing cloud based IAM services		
M3.01	Demonstrate securing AWS IAM	4	Applying
M3.02	Experiment with auditing AWS IAM	3	Applying
M3.03	Practice Secure AWS Directory Services	3	Applying
CO4	Demonstrate security monitoring and audit trails and securing storage services.		
M4.01	Implement Secure Amazon Simple Storage Service	3	Applying
M4.02	Experiment with securing Amazon Elastic Block Store	3	Applying
M4.03	Demonstrate security monitoring and audit trails using AWS CloudWatch	3	Applying
M4.04	Perform security monitoring and audit trails using AWS CloudTrail	3	Applying
	Lab Test – II	2	

Text / Reference

T/R	Book Title/Author
T1	Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance, Tim Mather, Subra Kumaraswamy, Shahed Latif, O'Reilly Media; 1st edition, 2009
T2	Cloud Security Handbook: Find out how to effectively secure cloud environments using AWS, Azure, and GCP, Eyal Estrin, Copyright © 2022 Packt Publishing

Online Resources

Sl.No	Website Link
1	https://docs.aws.amazon.com/
2	https://docs.microsoft.com/en-us/cli/azure.
3	https://cloud.google.com/sdk